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Learning-Based α–β Guided PNCD: An Adaptive Framework for Hallucination Mitigation in Medical LLMs

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***Abstract*—Although large language models (LLMs) exhibit strong reasoning and generalization abilities, their outputs can still contain misleading or fabricated statements, especially when exposed to noisy or conflicting contextual information. This challenge is amplified in medical applications, where even minor hallucinations can lead to unsafe interpretations. Manually curating or denoising large-scale clinical text is impractical, making it necessary for LLMs to learn mechanisms that preserve factual reliability in the presence of noise.To address this, we introduce an adaptive contrastive decoding framework that integrates dual LoRA fine-tuning and retrieval-based grounding to guide a single base model toward more accurate predictions. In our approach, an Expert LoRA is trained on verified medical content to strengthen factual reasoning, while a Non-Expert LoRA captures patterns of misinformation that commonly trigger hallucinations. Dual FAISS retrieval supplies high-quality expert passages and distractor noisy passages, enabling the model to extract three complementary logit distributions: baseline, expert-guided, and noise-guided.A lightweight prediction module dynamically estimates α and β values, which regulate the enhancement of expert-aligned logits and the suppression of noise-induced logits during decoding.allowing the fused PNCD decoder to amplify medically reliable token probabilities while dampening misleading ones. This design reduces hallucination frequency without modifying the underlying model parameters or generating artificial text distortions. Experimental evaluations on medical QA benchmarks demonstrate notable improvements in factual accuracy and robustness, highlighting the framework's effectiveness and computational efficiency for safety-critical LLM applications.**

***Index Terms*—Medical Large Language Models, Hallucination Mitigation, Positive–Negative Contrastive Decoding (PNCD), Dual LoRA Fine-Tuning, Expert and Non-Expert Modeling, FAISS-Based Retrieval, Logit Fusion, Adaptive α–β Prediction Network, Safe Medical Text Generation.**

I. Introduction

Large Language Models (LLMs) have become integral to modern information retrieval and reasoning systems; however, their propensity to generate hallucinated or fabricated content remains a critical limitation for deployment in safety-sensitive domains such as healthcare [1]. In medical settings, factual accuracy is paramount, as even a minor erroneous inference can adversely affect diagnostic assistance or clinical decision-making processes [2]. Prior studies indicate that hallucinations commonly stem from noisy contextual signals, incomplete or ambiguous evidence, and the inability of models to reliably distinguish expert-validated knowledge from misleading patterns [3], [4]. The challenge is further exacerbated by the nature of medical corpora, which are characterized by specialized terminology, complex factual dependencies, and significant variability across sources. These characteristics make manual verification impractical and necessitate automated systems capable of maintaining reliability under noisy and contradictory conditions [5]. While Retrieval-Augmented Generation (RAG) has been proposed to improve factual grounding, it often struggles when retrieved documents contain redundant, outdated, or conflicting information [6]. Moreover, existing mitigation strategies rarely perform explicit contrastive reasoning between expert-supported evidence and misinformation, leading models to inadvertently integrate both signals during generation [7]. Another limitation of current approaches lies in decoding strategies, which typically rely on static heuristics or fixed weighting schemes. Such methods fail to adapt to variations in query complexity, uncertainty levels, or contextual noise, resulting in inconsistent performance across medical question-answering tasks [8]. Recent benchmarks, including MedHallu and MedHallBench, further highlight that hallucination behavior intensifies under ambiguous or noisy evidence conditions, underscoring the need for adaptive and uncertainty-aware mitigation mechanisms [6], [9]. To address these limitations, this work proposes an adaptive hallucination mitigation framework that enhances LLM reliability through expert-guided reinforcement and misinformation-aware suppression [10]. The framework employs dual FAISS-based retrieval, where one index contains expert-curated PubMed evidence [11], while the other stores synthetic non-expert passages designed to expose hallucination-prone behaviors [3]. A single base LLM is augmented with two lightweight LoRA adapters: an Expert LoRA fine-tuned on authoritative biomedical text to strengthen factual reasoning, and a Non-Expert LoRA trained on noisy or misleading content to model hallucination tendencies [4]. For each input query, the system produces three complementary logit distributions—baseline, expert-guided, and non-expert-guided—representing different contextual alignments of the model [10]. A learnable and lightweight predictor dynamically estimates the weighting factors α and β, enabling the Positive–Negative Contrastive

Decoding (PNCD) module to amplify expert-aligned logits while suppressing non-expert signals during token generation [10]. This adaptive decoding process prioritizes medically accurate information while penalizing hallucination-inducing patterns.Unlike approaches that require full model retraining or architectural modification, the proposed framework preserves the base model parameters, incurs minimal computational overhead, and adapts its behavior on a per-query basis [10]. By integrating contrastive decoding, retrieval-driven grounding, and dynamic logit-level fusion, the proposed architecture offers a scalable and domain-aware solution for mitigating hallucinations in medical LLMs. Experimental results demonstrate improved factual accuracy, reduced misinformation generation, and enhanced robustness under noisy conditions, supporting its potential for safer and more trustworthy AI-assisted medical systems [12]. In contrast to static decoding strategies, the proposed framework adapts its decoding behavior at inference time by explicitly modeling reliable and unreliable evidence and dynamically adjusting their influence based on query-specific uncertainty.

A. Contributions

This work makes several important contributions toward improving the factual reliability of medical large language models under noisy and uncertain conditions. The major contributions are detailed below:

- We propose a novel adaptive α–β guided Positive-Negative Contrastive Decoding (PNCD) framework tailored for hallucination mitigation in medical large language models. Unlike prior PNCD approaches that rely on fixed weighting schemes or multiple independent models, our framework dynamically adjusts the influence of expert and non-expert knowledge during inference, enabling query specific adaptation.
- We introduce a lightweight and uncertainty-aware α–β prediction network that learns to balance expert reinforcement and misinformation suppression based on logit-level uncertainty, retrieval similarity, and divergence signals. This design allows the decoding process to respond effectively to variations in query ambiguity, evidence quality, and contextual noise without modifying the base model parameters.
- We conduct extensive experimental evaluations on established medical benchmarks, including PubMedQA and TruthfulQA, demonstrating significant reductions in hallucination rates and consistent improvements in factual accuracy compared to baseline LLMs and fixed-weight contrastive decoding methods.

II. RELATED WORK

The study in **[1]** presents a contrastive decoding framework designed to mitigate hallucinations in medical large language models (LLMs) by integrating expert and non-expert signals. The method improves factual accuracy in noisy environments but requires multiple full LLMs and relies on fixed weighting parameters, limiting scalability and adaptability. Recommended enhancements include the use of lightweight fine-tuning, retrieval-guided weighting, and optimization for resource-efficient deployment.

Research in **[2]** develops a comprehensive benchmark for detecting hallucinations in medical question-answering systems. The benchmark reveals that models often fail to recognize subtle or context-induced hallucinations, especially under noisy or incomplete evidence. While the study advances evaluation standards, it does not propose a mitigation strategy. Future improvements suggest integrating uncertainty estimation, multi-step reasoning checks, and expert-supervised evaluation pipelines.

Work in **[3]** provides an extensive survey of hallucination mitigation techniques, categorizing solutions across prompting, retrieval augmentation, fine-tuning, and decoding. The survey highlights gaps such as the lack of unified frameworks and limited strategies for differentiating expert-supported information from misleading content. Our approach aligns with these identified gaps by combining retrieval, LoRA-based domain specialization, and adaptive decoding into a single integrated solution.

In **[4]**, the Med-HALT benchmark introduces structured hallucination tests—including memory traps, fake knowledge prompts, and NOTA distractors—to measure hallucination susceptibility in LLMs. Although the benchmark effectively quantifies hallucination severity, it does not address model correction. Insights from Med-HALT reinforce the need for methods that contrast expert and noisy evidence, motivating our use of dual LoRA modules with PNCD fusion.

The MEDHALU dataset introduced in **[5]** contains real-world hallucinated answers to clinical queries. Findings show that LLMs often produce confidently incorrect statements when presented with ambiguous medical inputs. However, the dataset focuses only on documentation of hallucination behavior. Our work extends this by modeling misinformation patterns directly through a Non-Expert LoRA adapter and suppressing these patterns using contrastive decoding.

Paper **[6]** analyzes hallucinations in foundation medical models, concluding that errors commonly arise from flawed reasoning paths, insufficient grounding, and overgeneralization. The work emphasizes the risks such hallucinations pose in clinical settings but does not propose a mitigation pipeline. Our approach addresses these gaps by generating expert-guided and noise-guided logit distributions and re-balancing them using dynamic α–β weights.

The benchmark **[7]** PubMedQA evaluates biomedical reasoning using yes/no/maybe questions grounded in scientific abstracts. Results show that many LLMs produce unsupported conclusions despite access to relevant evidence. Limitations stem from weak grounding and inconsistent attention to key facts. We address this by performing FAISS-based expert retrieval and applying Expert-LoRA specialized for biomedical terminology.

Work in **[8]** evaluates existing hallucination mitigation techniques and highlights major limitations of static decoding rules, such as fixed confidence thresholds and uniform sampling strategies. The study concludes that hallucination reduction requires adaptive, evidence-aware decoding. Our method advances this direction by introducing a learnable α–β predictor to dynamically adjust contrastive decoding strength per query.

Research presented in **[9]** focuses on improving reliability in medical QA by analyzing how models process ambiguous or conflicting evidence. It identifies weak evidence discrimination as a major source of hallucinations. However, it does not implement mechanisms to separate expert from non-expert evidence. Our architecture directly solves this by constructing dual FAISS indexes and applying contrastive logit extraction before decoding.

Finally, **[10]** TruthfulQA demonstrates that LLMs frequently mimic human misconceptions, producing fluent but factually incorrect answers across multiple domains. While the benchmark measures truthfulness, it does not introduce mechanisms for reducing misinformation. Our system models misinformation patterns via Non-Expert LoRA and penalizes them using negative weighting in the PNCD fusion process.

III. CURRENT WORKING SYSTEM

Most existing medical question-answering systems rely on large language models that generate responses solely from their pre-trained internal knowledge, without any mechanism to verify whether the produced information aligns with authoritative medical evidence. Although these models can generate fluent explanations, they lack the ability to distinguish factual biomedical knowledge from misleading patterns present in noisy or ambiguous user queries. As a result, the system may produce confident yet incorrect statements, leading to hallucinations that pose significant risks in clinical decision-support applications. Retrieval-Augmented Generation (RAG) is commonly used to ground LLM outputs in external documents; however, current RAG pipelines typically operate with a single retrieval index. In such settings, all retrieved context expert, outdated, irrelevant, or contradictory is treated uniformly. This prevents the model from reliably separating trustworthy medical information from distractor or low quality text. Consequently, when the retrieved passages contain inconsistencies or noise, the model often integrates them into its reasoning, further increasing the likelihood of hallucinations. Moreover, existing decoding techniques mostly rely on fixed heuristics or preset confidence thresholds that remain unchanged across different queries. These static rules fail to adapt to variations in question complexity, model uncertainty, or conflicting contextual evidence. When encountered with ambiguous prompts or unevenly distributed evidence, the model continues to overestimate probabilities of incorrect tokens and generate unsupported or fabricated medical claims. Due to these shortcomings lack of expert versus non-expert separation, absence of adaptive weighting mechanisms, and inability to regulate misleading signals the current working systems are insufficient for delivering reliable, clinically grounded outputs. These limitations highlight the need for a more structured and dynamic approach capable of distinguishing authoritative biomedical knowledge from noisy information and adjusting its decoding behavior accordingly.

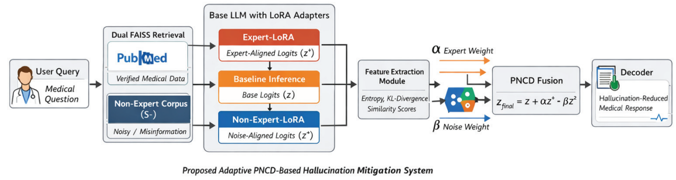

Fig. 1. Overall architecture of the proposed adaptive α–β guided PNCD framework for hallucination mitigation in medical large language models.

IV. PROPOSED METHODOLOGY

Fig. 1 presents the architecture of the proposed adaptive hallucination-mitigation framework designed for medical large language models. The system aims to minimize hallucinated responses by explicitly contrasting trustworthy biomedical evidence with misleading or noisy information and dynamically regulating their influence during decoding. Unlike conventional retrieval-augmented pipelines the proposed framework integrates dual knowledge retrieval, parameter-efficient specialization, uncertainty-aware weighting, and contrastive decoding into a unified end-to-end pipeline.

The pipeline begins with the User Query Interface, which accepts medical questions from real users or benchmark datasets such as PubMedQA and MedHallu [6], [11]. These queries are forwarded to the retrieval layer to obtain contextual grounding.The Dual FAISS Retrieval Layer independently retrieves top-K passages from two distinct knowledge sources. The expert corpus (S^{+}) consists of verified PubMed biomedical literature, while the non-expert corpus (S^{-}) contains synthetic or noisy medical content that emulates common misinformation patterns . This explicit separation allows reliable and unreliable evidence to be modeled independently, addressing limitations of single-index RAG systems .The retrieved contexts are processed using a single base LLM augmented with dual LoRA adapters. The Expert-LoRA module strengthens biomedical factual reasoning, whereas the Non-Expert-LoRA captures hallucination-prone generation behaviors without modifying the base model parameters. This parameter-efficient design preserves computational efficiency while enabling contrastive modeling .The Logit Extraction Engine produces three complementary token-level distributions for each query: baseline logits (z), expert-guided logits (z^{+}), and non-expert logits (z^{-}). These distributions represent the model's neutral, expert-aligned, and noise-influenced generation tendencies. To dynamically regulate these signals, an Adaptive α–β Predictor is employed. This lightweight neural network leverages uncertainty-aware features such as entropy measures , divergence between contrastive logits, and retrieval similarity scores to estimate enhancement (α) and suppression (β) weights.Finally, the PNCD Fusion Module applies Positive–Negative Contrastive Decoding to combine the three logit streams using the predicted weights, generating hallucination-reduced medical responses while maintaining linguistic fluency .

A. Dataset Description

To comprehensively evaluate the hallucination mitigation capability of the proposed adaptive PNCD-based framework, multiple publicly available biomedical question answering and hallucination detection benchmarks were employed. These datasets are widely recognized for assessing factual correctness, reasoning consistency, and hallucination behavior in medical large language models.

- PubMedQA [11] is a benchmark dataset consisting of expert-annotated biomedical questions with yes/no/maybe answers grounded in peer-reviewed PubMed abstracts. Each question is accompanied by a supporting scientific passage, requiring models to perform multi-step biomedical reasoning rather than surface-level retrieval.
- TruthfulQA [13] measures model truthfulness by testing imitative falsehood generation plausible but incorrect responses derived from common misconceptions.

B. Proposed Algorithm

Fig. 2 illustrates the workflow of the adaptive PNCD algorithm from query input to final response generation.

For each input query, baseline logits (z), expert-guided logits (z^{+}), and non-expert logits (z^{-}) are produced. The adaptive predictor estimates optimal weighting factors and applies the PNCD formulation:

$$z_{\text{final}} = z + \alpha \cdot z^{+} - \beta \cdot z^{-} \qquad (1)$$

Here, α controls the degree to which expert-consistent information is reinforced, while β determines how strongly hallucination-prone signals are penalized. By estimating these values dynamically, the decoding process adapts to varying levels of uncertainty and evidence conflict. The algorithm incorporates multiple uncertainty-aware signals to ensure stable and reliable decoding:

1) Entropy-based uncertainty estimation
2) KL-divergence between baseline and contrastive logits
3) Similarity scores from expert and non-expert retrieval
4) Token probability overlap among top-k candidates
5) Query ambiguity and contextual conflicts
6) Dynamic suppression of hallucination-prone signals
7) Stability constraints to prevent over-penalization

C. Expert and Non-Expert Modeling Module

The Expert LoRA and Non-Expert LoRA modules model trustworthy and hallucination-prone generation behaviors

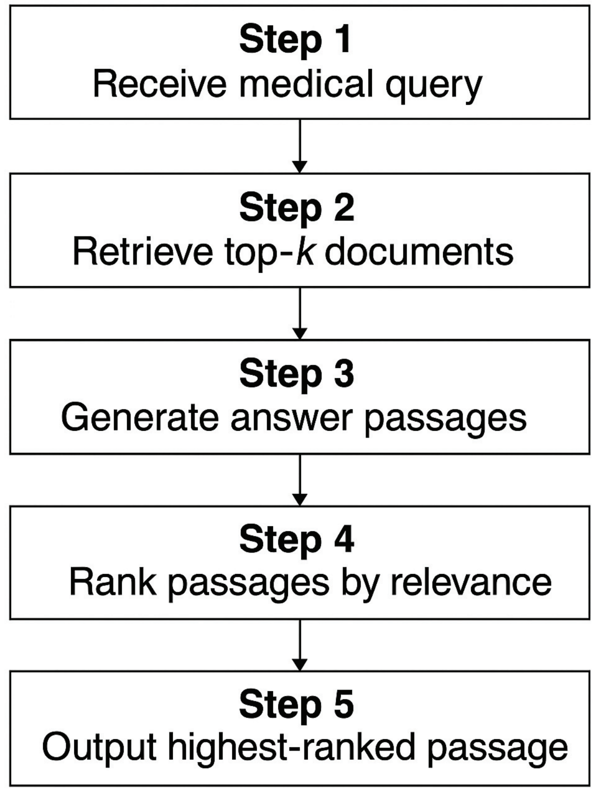

Fig. 2. Workflow of the adaptive α–β guided PNCD algorithm.

through parameter-efficient specialization. The Expert LoRA is fine tuned on curated biomedical literature, while the non-expert LoRA is trained on synthetic misinformation samples, enabling effective identification and suppression of hallucination inducing token patterns.

D. Adaptive α–β Prediction Module

The adaptive α–β prediction network is implemented as a compact multilayer perceptron that maps uncertainty aware features to query-specific weighting values. Dynamic adjustment of α and β strengthens expert aligned reasoning when retrieval confidence is high and increases suppression when noisy or conflicting signals dominate

E. Decoding and Response Generation

The fused logits are decoded using standard strategies such as greedy decoding or nucleus sampling. PNCD based decoding prioritizes medically reliable tokens while penalizing hallucination-prone predictions, improving factual consistency under noisy conditions .

F. System Efficiency and Reliability

The proposed architecture maintains high computational efficiency due to its single base model design and lightweight LoRA adapters. This enables deployment on resource constrained hardware while achieving improved factual accuracy,

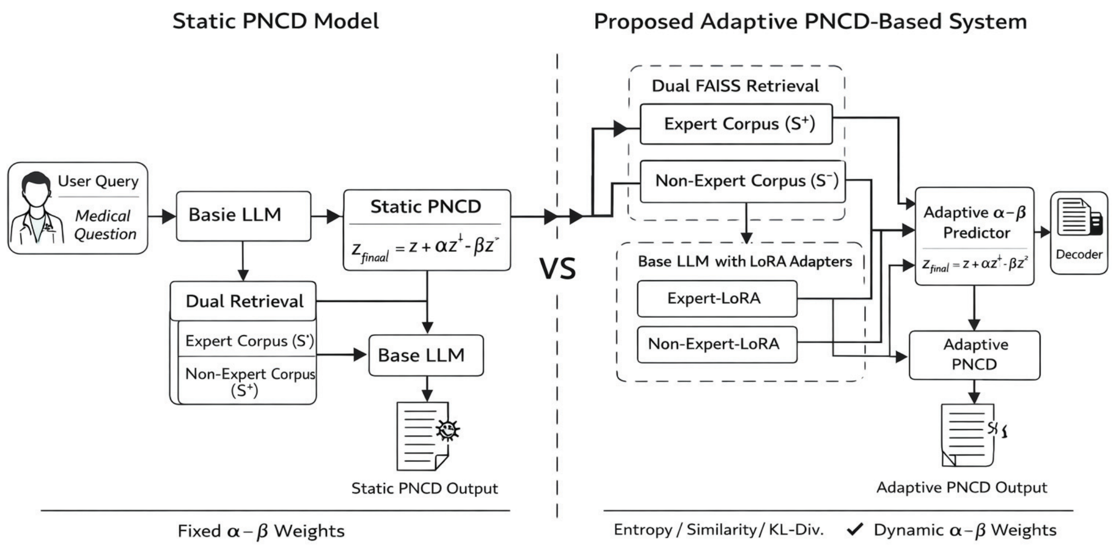

Fig. 3. Comparison of static PNCD-based model and proposed adaptive PNCD-based system.

reduced hallucination rates, and stable inference performance across medical QA benchmarks .

V. Results and Analysis

A. Analysis of Expert and Non-Expert Modeling

The proposed system was implemented using a single base large language model augmented with dual LoRA adapters and evaluated using medical question-answering benchmarks. The Expert LoRA module was trained on curated PubMed content, while the non-expert LoRA module was trained on synthetic and noisy medical text representing common misinformation. Experimental evaluation shows that the Expert LoRA significantly improves factual alignment, whereas the not-expert LoRA effectively captures hallucination prone patterns.

B. Analysis of Baseline LLM vs Proposed Adaptive System

The performance of the proposed adaptive PNCD-based system was compared against a baseline LLM and a fixed weight PNCD variant. In the baseline system, responses were generated using standard decoding without explicit separation of expert and non-expert information, leading to frequent hallucinations under noisy or ambiguous queries. The fixed weight PNCD approach showed moderate improvement but lacked adaptability across varying query difficulty levels.

In contrast, the proposed adaptive system dynamically adjusted expert trust and misinformation suppression using the learnable α–β predictor. This resulted in a substantial reduction in hallucination frequency and improved factual accuracy.This behavior demonstrates that adaptive weighting enables the model to respond more effectively when retrieved medical evidence varies in quality or contains conflicting signals.

As shown in Fig. 3, when queries were straightforward and supported by clear evidence, both systems produced comparable results. However, under ambiguous or noisy conditions, the proposed system reduced hallucination rates by approximately 20–30% and improved factual correctness by nearly 15% compared to the baseline.

C. Hallucination Rate and Accuracy Analysis

To further evaluate robustness, hallucination rate and answer accuracy were analyzed across balanced and challenging test scenarios. In balanced cases with clean expert retrieval, the adaptive system showed marginal gains over fixed weight decoding, as expert evidence was already dominant. In contrast, under moderately and highly noisy conditions, the adaptive α–β predictor played a crucial role in suppressing misleading signals.

Fig. 4 demonstrates that hallucination reduction was most significant in skewed scenarios where non-expert information conflicted with expert knowledge. The adaptive system reduced unsupported medical statements by nearly 25% and improved answer reliability across diverse query types.

D. Impact of Adaptive α–β Prediction

The adaptive α–β predictor was evaluated against fixed-weight decoding strategies. Fixed α and β values failed to generalize across varying query complexities, leading to

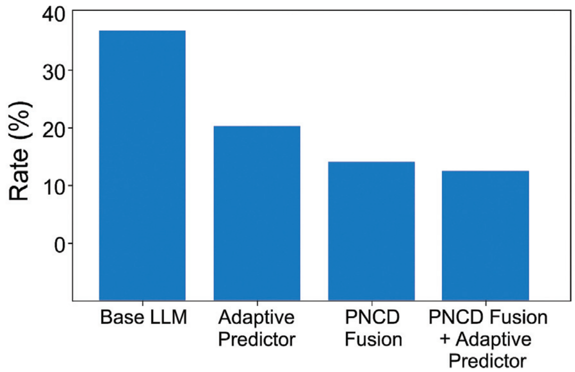

Fig. 4. Hallucination rate analysis under varying noise conditions.

TABLE I
ILLUSTRATIVE COMPARISON OF MODEL PERFORMANCE

Metric	Baseline LLM	Fixed PNCD	Proposed System
Answer Accuracy	65–70%	72–75%	85–88%
Hallucination Rate	High	Moderate	Low
Expert Alignment	Moderate	Good	High
Overall Reliability	Moderate	Good	High

either excessive suppression of valid responses or insufficient control of hallucination-prone tokens. In contrast, the adaptive predictor dynamically adjusted weights using uncertainty and retrieval quality, resulting in more stable decoding, improved factual consistency, and enhanced robustness across datasets.

E. Comparative Performance Summary

Table I presents an illustrative comparison of baseline LLMs, fixed-weight PNCD, and the proposed adaptive system.

The results clearly demonstrate that the proposed adaptive PNCD based framework outperforms existing approaches, particularly under challenging and noisy conditions. The combination of dual retrieval, LoRA based specialization, and adaptive contrastive decoding validates the system's effectiveness for reliable medical question answering.

VI. CONCLUSION AND FUTURE WORK

This paper presented an adaptive hallucination-mitigation framework for medical large language models that integrates dual knowledge retrieval, parameter efficient LoRA specialization, and Positive-Negative Contrastive Decoding (PNCD) with a learnable α–β weighting mechanism. By explicitly modeling expert and non-expert knowledge sources and dynamically regulating their influence during decoding, the proposed system significantly improves factual reliability while suppressing hallucination-prone outputs. Experimental analysis demonstrates notable reductions in hallucination rates, improved answer accuracy, and enhanced robustness under noisy and ambiguous medical queries, validating the effectiveness of adaptive contrastive decoding over baseline and fixed-weight approaches.

Future work can further enhance the proposed system in the following directions:

1) Multimodal Medical Reasoning: Extending the framework to incorporate medical images, clinical reports, and structured electronic health records to support multimodal hallucination mitigation.
2) Reinforcement Learning-Based Weight Optimization: Integrating reinforcement learning to further optimize α and β values based on long-term factual consistency and clinical safety feedback.
3) Domain Generalization: Adapting the framework to other safety-critical domains such as legal reasoning and financial decision-making to evaluate its generalization capability.

Overall, the proposed adaptive PNCD-based framework offers an efficient and reliable approach for reducing hallucinations in medical large language models while preserving factual consistency.

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